Homework Flow — Cost Verification & Provider Reconciliation

Purpose. This document lets us *confirm* the per-session cost the estimator shows for the Homework Help (Tier 2) flow. It checks every provider unit price against the live provider documentation, then shows exactly how the two biggest lines — **TTS audio** and **real-time image generation** — are counted, and explains the o3 reasoning-token risk.

Bottom line up front: the unit prices are *correct and current*. The headline number is driven by three deliberate, conservative choices: **audio is billed on the MAXIMUM narration length** (Section 2), **a diagram image is generated in real time at every walkthrough step of every session** (Section 3), and every price carries a **+50% buffer**. The figure to keep validating on a real invoice is o3's **hidden reasoning tokens** (Section 4.3).

1. Provider unit prices — confirmed against documentation (July 2026)

Service	Model	Published list price	In estimator	Confirmed
Reasoning LLM	**o3**	**$2.00 / 1M input · $8.00 / 1M output**	$2 / $8	✅ matches OpenAI
Prose / vision LLM	**gpt-4.1**	$2.00 / 1M input · $8.00 / 1M output	$2 / $8	✅ matches OpenAI
Diagram image	**Gemini 3.1 Flash Image**	**$0.067 / image (standard tier)**	$0.067	✅ as configured
Text-to-speech	**Inworld TTS-2**	**$25.00 / 1M characters (on-demand)**	$25	✅ matches Inworld
Safety	omni-moderation	Free	$0	✅

The +50% safety buffer. Every price above is multiplied by **1.5** inside the estimator before any figure is shown, as an internal margin for retries, prompt drift and future price moves. So the effective rates the estimator charges against are: o3 / gpt-4.1 **$3 / $12** per 1M, Gemini **$0.1005** per image, Inworld **$37.50** per 1M characters. Throughout, we show **both** columns — "*list*" (the provider's published price, for reconciliation) and "*+50%*" (what the estimator displays).

---

2. Text-to-speech (Inworld TTS-2) — billed on the MAXIMUM narration length

Inworld bills **per character** sent to synthesis. Real homework narration runs far longer than a "typical" estimate, so the estimator costs audio on the **maximum** a session can produce — every audio driver at its ceiling. This is the figure shown on the dashboard.

Phase narrated	Max characters	At the slider ceiling
3A explanation	800	`charsExp` max
Solution + parallel-demo steps	10,000	`10 steps × 500 × 2`
Intros + completion	1,200	`charsOv` max
Attempt feedback + Q&A	12,000	`(12 + 8) × 600`
Maximum per session (billed)	**24,000**	

Cost of 24,000 characters: list `24,000 ÷ 1,000,000 × $25` = **$0.60** · +50% = **$0.90**.

Inworld gets cheaper with commitment, and we estimate on the most expensive on-demand tier, so a real audio invoice can only come in lower:

Inworld TTS-2 tier	$ / 1M characters	Cost of a 24,000-char session
On-demand (what we estimate on)	**$25.00**	$0.60
$300 / month commit	$15.00	$0.36
$1,500 / month commit	$12.50	$0.30
Enterprise	$10.00 (as low as $5)	$0.24 (↓ $0.12)

---

3. Image generation (Gemini 3.1 Flash Image) — one per step, every session

Diagram images are generated **in real time at every displayed walkthrough step of every session** — this is *not* gated on whether the child's upload contained a figure. One image is produced for the 3A explanation, one for each parallel-demo step, and one for each "your-turn" step:

```
images / session = 1 (explanation)
                 + avgSteps (demo steps)
                 + avgSteps (your-turn steps)
                 = 2 × avgSteps + 1
```

At the default **5 steps**, that is **11 images per session**. Because it scales with walkthrough depth, it grows with longer walkthroughs:

Walkthrough depth (steps)	Images / session ( )	Cost (list @ $0.067)	Cost (+50%)
3	7	$0.469	$0.704
5 (default)	**11**	**$0.737**	**$1.106**
7	15	$1.005	$1.508
10 (max)	21	$1.407	$2.111

> **This is now the single largest line in a session.** At the default depth, image generation is **$0.74 list / $1.11 buffered** — more than audio. If in practice not every step renders a fresh image (e.g. some are cached or reused), lower the step-image assumption and this line drops proportionally.

---

4. o3 reasoning model — tokens and cost

o3 is only used for **calculation / derivation** questions (Maths, Physics, Chemistry, computational Science). Prose/explanation questions run on gpt-4.1. The build pipeline is five LLM calls (steps A · parallel B · demo C · 3A explanation · answer key D); A→B→C can regenerate on a self-check failure, so they carry a retry factor (default **1.15×**).

4.1 Per-call token assumptions (o3 route)

Output tokens below are set to **include** o3's hidden reasoning tokens — see Section 4.3.

Call	Input tok	o3 output tok	Runs / session	o3 cost (list)
A · Solution steps	700	3,000	1.15	$0.0292
B · Parallel question	800	2,500	1.15	$0.0248
C · Parallel demo (worked)	900	3,000	1.15	$0.0297
3A · Explanation	600	2,200	1.00	$0.0188
D · Private answer key	900	3,000	1.00	$0.0258
Validate attempt	700	1,500	3.00	$0.0402
Follow-up Q&A	800	1,500	1.00	$0.0136
o3 subtotal	**~7,160**	**~20,975**		**$0.1821**

Plus the fixed gpt-4.1 intake/classify calls that run in **every** session (image extract + analyze + match): **$0.0118**.

4.2 Per-session o3 cost (the math)

```
o3 input :  7,160 tok ÷ 1,000,000 × $2  = $0.0143
o3 output: 20,975 tok ÷ 1,000,000 × $8  = $0.1678
o3 total (list)                          = $0.1821  →  +50% = $0.2732
```

4.3 Why o3 can look cheaper than the docs — hidden reasoning tokens

The unit price is right ($2/$8); the risk is the token count, not the rate. o3 bills *invisible* reasoning tokens at the **output** rate. OpenAI's documentation notes a single o3 answer can burn **5,000–20,000 internal reasoning tokens**. Our model already assumes ~21,000 o3 output tokens for a whole calculation session (near the high end), but if o3 reasons *harder*, cost scales. This table shows a full calculation session (including the max-audio $0.60 and the 11-image $0.737 lines) as reasoning load rises:

o3 reasoning load	o3 output tok / session	Calc session (list)	Calc session (+50%)
As modelled	~21,000	$1.531	**$2.297**
1.5× harder	~31,500	$1.615	$2.423
2× harder	~42,000	$1.699	$2.549
3× harder	~63,000	$1.867	$2.801

> The o3 output-token counts are **editable** in the estimator's *Advanced assumptions* — tune them to a sample of real billed sessions and the estimate tracks reality.

5. Full homework-session confirmation table

Per-session cost by service, with **audio at the 24,000-char maximum** and **11 real-time images** (default 5 steps). **Calculation** sessions use o3; **explanation** sessions use gpt-4.1; the **blended** row is the estimator default (50/50, image intake, audio on).

Session type	o3	gpt-4.1	Gemini images (×11)	Inworld audio (max)	Total (list)	Total (+50%)
Calculation (o3)	$0.182	$0.012	$0.737	$0.600	**$1.531**	**$2.297**
Explanation (gpt-4.1)	—	$0.070	$0.737	$0.600	**$1.407**	**$2.111**
Blended (default)	$0.091	$0.041	$0.737	$0.600	**$1.469**	**$2.204**

Every number is reproducible from Sections 2–4 with the published unit prices — that is the confirmation: **provider price × quantity = the estimator's figure.** Image generation and audio together are ~85% of a session, because both are costed at their conservative maximum.

6. Scaling to volume (monthly / yearly)

Using the **blended buffered** cost of **$2.204** per session, scaled by each preset's monthly session count (the *range* uses the explanation-only and calculation-only bounds, $2.111 → $2.297):

Preset	Students × questions	Sessions / month	Monthly (blended)	Yearly (blended)	Monthly range
Reset to defaults	200 × 10	2,000	**$4,408**	**$52,896**	$4,222 – $4,594
Light usage	300 × 20	6,000	**$13,224**	**$158,688**	$12,666 – $13,782
Heavy usage	600 × 40	24,000	**$52,896**	**$634,752**	$50,664 – $55,128

Monthly figures use the blended-default per-session cost; the Light/Heavy presets also raise walkthrough depth (more steps → more images and narration), so their true per-session cost is a little higher — treat these as a lower-middle estimate. The two levers that move these most: the step-image assumption (Section 3 — fewer images per step if some are cached) and the Inworld tier (a $1,500/mo commit halves the audio line).

---

7. Reconciliation summary

1. **Unit prices are correct and current** (verified July 2026) — o3 & gpt-4.1 at $2/$8 per 1M tokens, Gemini image at $0.067, Inworld TTS-2 at $25 per 1M characters on-demand.
2. **Image generation is now the largest line** — a real-time diagram at every walkthrough step of every session (`2×steps + 1` ≈ 11 images at default depth) = **$0.74 list / $1.11 buffered**. This reflects that images are created live at each phase, not once per upload.
3. **Audio is billed at the maximum narration length** (24,000 chars ≈ $0.60 / $0.90).
4. **Everything carries a +50% buffer** and audio is priced on Inworld's most expensive on-demand tier, so the estimate is deliberately conservative.
5. **Validate two things on real usage:** (a) whether every step truly generates a fresh image or some are cached/reused — this is the biggest single assumption; and (b) o3's reasoning-token count (Section 4.3).

Recommended checks before committing to a budget

- Count the **actual images generated** in a sample of real sessions vs. the `2×steps + 1` assumption; if some steps reuse a cached image, lower the assumption and the (dominant) image line falls proportionally.
- Confirm the **maximum narrated characters** per session vs. the 24,000-char ceiling.
- Pull the **actual o3 output-token count** from a dozen real calculation sessions vs. ~21,000/session.
- Decide the Inworld **commitment tier** (on-demand $25 → $1,500/mo commit $12.50 halves audio).

Sources

- OpenAI o3 API pricing ($2 / $8 per 1M tokens; hidden reasoning tokens billed at output rate): https://developers.openai.com/api/docs/pricing · https://pricepertoken.com/pricing-page/model/openai-o3
- Inworld TTS-2 pricing ($25 / 1M characters on-demand, tiered to $10 enterprise): https://inworld.ai/pricing · https://www.buildmvpfast.com/api-costs/ai-voice

Figures are grounded in the estimator's editable token / character / image assumptions; tune those to your real provider contracts and billed usage.